

Validated Turnout Against Self-Reported Turnout

Counted ballots against what people said they did, 1964–2024

Democracy's Data

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Almost every published claim about who takes part in American elections rests on a survey. The Census Bureau asks about 60,000 households after each federal election whether they voted, and nobody checks the answers.¹ The states, separately, count the ballots.

That makes turnout a rare thing: a survey question with an outside answer key. Turnout measured by asking is called **self-reported**; turnout checked against a record of ballots actually cast is **validated**, and the second is the standard the first is held to here. Do the two numbers agree? The question matters to a citizen because the survey is the only source that reports turnout by race, age and education. Those figures are quoted in voting-rights litigation and in nearly every argument about unequal participation. If the survey is wrong it is wrong in a direction, and the published gaps between groups are wrong with it.

¹ The Current Population Survey interviews “a probability selected sample of about 60,000 occupied households” every month; the voting questions are added in November of even-numbered years. U.S. Census Bureau, *Current Population Survey: Methodology*, <https://www.census.gov/programs-surveys/cps/technical-documentation/methodology.html>.

01 The test

This brief joins one source from each of two earlier sections of this book. The survey half is the U.S. Census Bureau’s Current Population Survey *Voting and Registration Supplement*, historical Table A-1, covering November 1964 through 2024 (<https://www.census.gov/data/tables/time-series/demo/voting-and-registration/voting-historical-time-series.html>). The record half is the count of ballots cast for president in the same 16 elections, from Jason Timm’s *PresElectionResults*, which compiles the official state canvasses (<https://github.com/jaytim/PresElectionResults>).

Those two sources can answer the question because they measure the same event and share nothing else. The Bureau’s interviewers never see a ballot, and the state canvassers never see the survey. Their only common column is the election year, and that is the key the two files join on. A year is about the best-behaved join key in this book. Join Keys and Crosswalks is the chapter on what a join key can and cannot promise. It also covers the crosswalk, a table that translates one set of identifiers into another when two files share no key at all.

One thing about the record half is worth saying before the numbers start. No federal agency records a national popular vote, because the United States does not hold one. The counted-ballot column here is a compilation of the state canvasses, assembled by people rather than published by a government. The word “actual” does slightly more work than the record behind it pays for.

02 The data

The table is `cps_turnout.csv`: one row per presidential election, 16 rows from 1964 through 2024. `year` is the join key. `vap_thousands` is the voting-age population the Bureau published, `citizen_vap` the same population counting citizens only, and `pct_citizen_basis` the share of citizens who said they voted. `reported_voters` is how many people the survey implies said they voted, `actual_votes` the ballots cast for president. `cps_rate`, `actual_rate` and `over_report_pp` hold the two rates and the distance between them. Four columns carry reported turnout by group: `pct_white_nh`, `pct_black`, `pct_asian` and `pct_hispanic`.

Year	Voting-age population	% voted, citizen basis	Said they voted	Citizen VAP	Ballots for president
2016	245,502,000	61.4	137,481,120	223,910,619	128,838,342

Read that one row as the join reads it. In 2016 the Bureau put the number of citizens of voting age at 223,910,619, and 61.4% of them told the survey they had voted, which comes to 137,481,120 people. The same year the states counted 128,838,342 ballots for president. Set one against the other and the survey's electorate is the larger by 8,642,778 people, 6.7% more voters than ballots. Every point in the figures below is that subtraction, repeated for 16 elections.

The Bureau publishes the survey half as a spreadsheet laid out for a person to read on a page, its headings spread over four rows; the figures used here are the *Total, Percent Voted* line of that table, one for each election.

Three things about how the table was put together settle what it can be asked.

The count of self-reported voters is derived, not published. Table A-1 prints percentages and one population total. `reported_voters` is the citizen voting-age population multiplied by the published percentage of citizens who said they voted (before 1980, when the Bureau published no citizen figures, the whole voting-age population and its percentage), which is arithmetic on the Bureau's own figures rather than an estimate of anybody's.

Ballots for president are not the whole electorate. People vote and skip the top of the ticket, so the true ballot count is a little higher than the column used here. Every gap below is a small overstatement for that reason.

The rate columns are empty before 1980. The Bureau began publishing a citizen basis only when the difference between residents and citizens grew large enough to matter. Quietly dropping those blank rows would cut the series' first four elections, among them 1972, the year the gap was proportionally widest.

Before you read on, commit to a direction. Over the six decades in this file, does the gap between what people say and what the ballots show get **larger**, stay **flat**, or get **smaller**? Write down one of the three.

03 What it says about democracy

More people say they voted than there are ballots, in 15 of 16 elections. The average excess is 6.8 million people, and in 2016 it was 8,642,778. That is the finding at its plainest: the survey describes an electorate the states never counted.

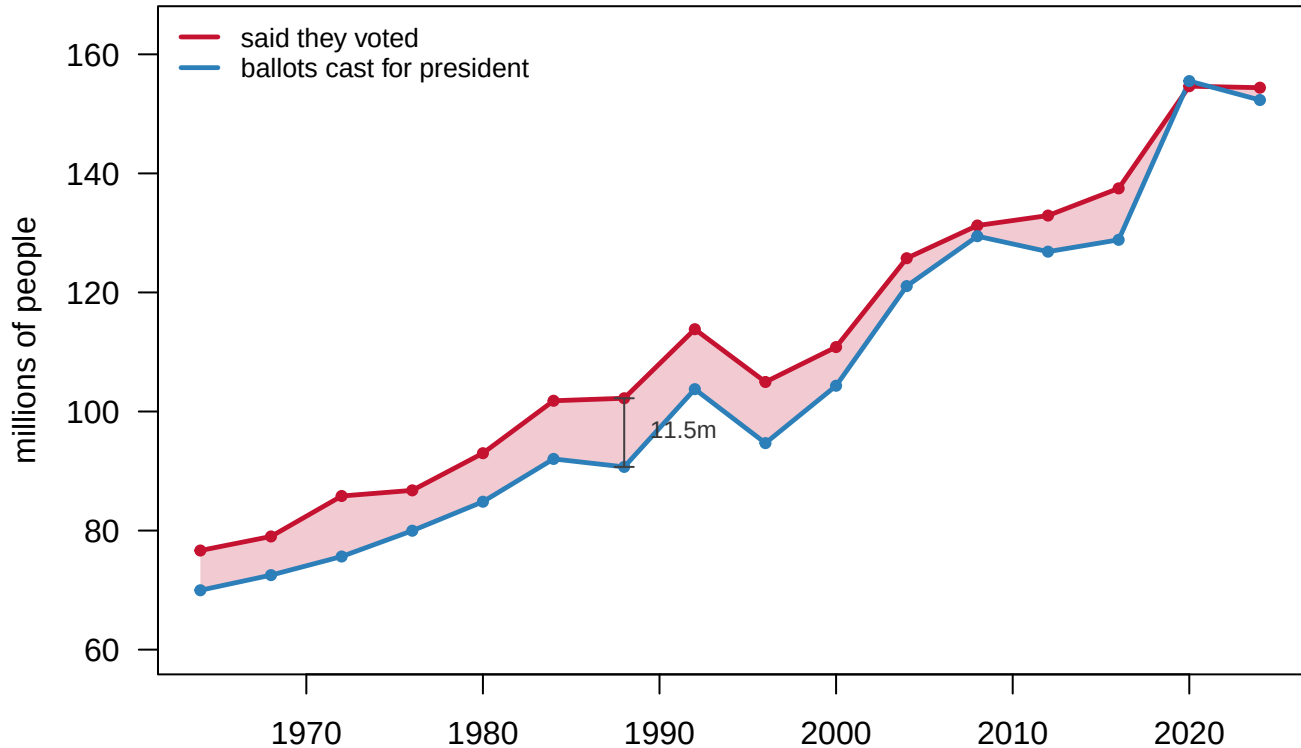


Figure 1. What people said, and what was counted, 1964 to 2024. The upper line is how many Americans the survey implies said they voted; the lower line is ballots cast for president, and the shaded wedge between them is the gap. Two lines with the space between them filled in is the right form here because the question is about a difference rather than a level. Both series roughly doubled over sixty years; the wedge did not. The widest year is 1988, at 11.5 million people.

The obvious explanation arrives first and almost everybody writes it down: people lie about voting. It is not an absurd reading. Voting is socially desirable, the interviewer represents the federal government, and admitting you stayed home is mildly embarrassing. Over-reporting of that kind is real and well documented.²

But notice what the explanation predicts. If social pressure drives the gap, the gap should hold roughly steady, or grow as voting becomes more moralised in a polarised era. It does neither.

² Stephen Ansolabehere and Eitan Hersh, "Validation: What Big Data Reveal About Survey Misreporting and the Real Electorate," *Political Analysis* 20, no. 4 (2012): 437–459, <https://doi.org/10.1093/pan/mps023>.

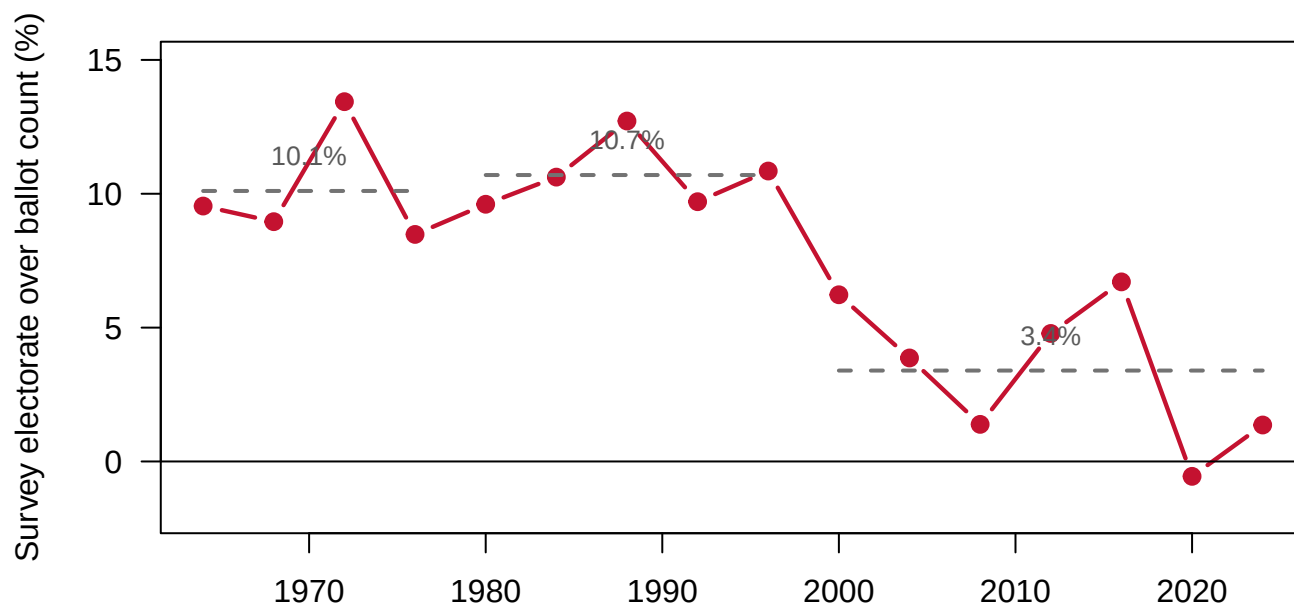


Figure 2. The gap, election by election, 1964 to 2024. The vertical axis is how far the survey electorate exceeded the ballot count, as a percentage of ballots. The dashed segments are era averages: 10.1% through the 1960s and 1970s, 10.7% through the 1980s and 1990s, and 3.4% since 2000.

The gap shrank by roughly three quarters, and nobody repaired it. There was no reform, no announced correction, no working group. Measured in percentage points of turnout, the survey overstated the rate by 7.0 points in 1988 and by 0.88 points in 2024.

The exceptions are informative rather than noisy. 2020 is the only election in the file where the survey came in *below* the ballot count, by 863,514 people. It is also the highest-turnout election in the series, which is what you would expect if part of the gap is people who did not vote being under-represented among those who answer surveys. The largest proportional gap, 13.4% in 1972, sits at the other end of the same story.

That decline breaks the explanation everyone reaches for first. Human vanity did not fall by three quarters between 1972 and 2020. Whatever the gap is made of, a constant of human nature is not it.

Now the part that is actually used. The Bureau's by-group table is the source of nearly every published claim about unequal participation, and every line in it is a self-report.

Year	White (non-Hispanic)	Black	Asian	Hispanic
2024	70.5	59.6	57.1	50.6
2020	70.9	62.6	59.7	53.7
2016	65.3	59.4	49.0	47.6
2012	64.1	66.2	47.3	48.0
2008	66.1	64.7	47.6	49.9

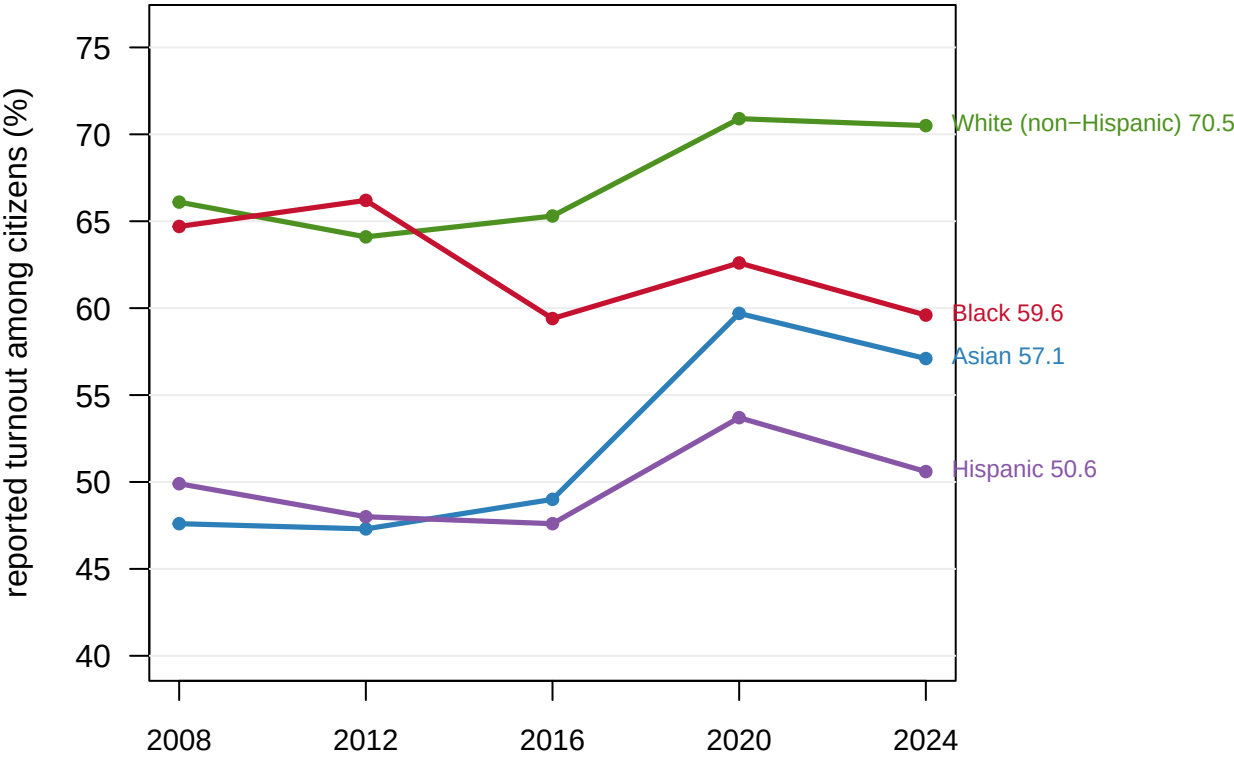


Figure 3. Reported turnout by race, five elections. One line per group, as the Bureau publishes them: the share of citizens in each group who told the survey they voted. The spread between the highest and lowest group was 18.5 points in 2008 and 19.9 points in 2024.

In 2024 reported turnout ran from 70.5% among White non-Hispanic citizens to 50.6% among Hispanic citizens. Everything established above applies to each of those numbers, and applies **unequally**. There is no reason to assume the over-report is the same size in every group, and if it is not, the gaps between the columns are wrong by an unknown amount in an unknown direction.

So the answer to the opening question depends on which instrument you trust. The survey says one thing about the size of the American electorate and the record says another, and the two disagree by millions of people in most years. Which one you reach for decides the answer, and only one of them can be checked.

04 What this brief cannot tell you

It cannot say why the gap exists. At least three mechanisms live inside one difference, and the file separates none of them.

Mechanism	What it is	Which way	Can we see it
Over-claiming	Respondents misremember, or say what they think they should have done	inflates the gap	No
Who answers at all	The people who agree to a government interview are unusually civic	either direction	No

Mechanism	What it is	Which way	Can we see it
Our own construction	Derived voter counts, and ballots for president rather than all ballots	inflates the gap	Partly

Consider the second row seriously. Survey response rates fell across American household surveys over exactly the period the gap shrank.³ If the people who still answer a government survey are disproportionately voters, the raw sample drifts toward voters on its own. The Bureau's weighting, which means counting some answers more than others so the sample matches the country, corrects for age, sex, race and region. It does not correct for how civic-minded somebody is, because nobody holds a population total for that.

A fourth mechanism sits inside the Bureau's arithmetic rather than in anybody's answer. A person who is interviewed but does not answer the voting question — says they do not know, refuses, or is never asked — is counted by the Bureau as a non-voter. Hur and Achen (2013) showed that this coding rule, as the share of such non-answers grew, pulled the published rate down election by election without anyone lying less.⁴ Part of the shrinking wedge in Figure 2 may be that rule doing its quiet work.

It cannot tell you whether validation studies measure lying or measure record linkage — whether a mismatch is a person who lied or a person whose line in the voter file was never found. Ansolabehere and Hersh (2012) matched a large survey to a commercial voter file and concluded that self-reports are badly inflated. Berent, Krosnick and Lupia (2016) answered that a failed match looks exactly like a liar, because names change, addresses change and records get purged.⁵ Every audit needs its own instrument, so the data cannot arbitrate.

It cannot give the by-group figures a margin of error. The published table carries nothing saying how far off each percentage could be, and the Asian and Hispanic figures rest on the smallest subsamples in it.

And it covers presidential years only. Midterms are absent from this file, and there is every reason to expect the gap behaves differently in them.

05 What you should have learned

- Self-reported turnout has exceeded the counted ballots in 15 of 16 presidential elections, by 6.8 million people on average and by 8,642,778 in 2016.
- The excess fell from about 10.1% of the ballot count in the 1960s and 1970s to about 3.4% since 2000, with no correction announced by anybody, which rules out any explanation that is a constant of human nature.
- 2020 is the one election where the survey came in below the count, and it is also the highest-turnout election in the file.
- Both halves of this table are constructed: `reported_voters` is a percentage multiplied by a population, and `actual_votes` is a compilation of state canvasses rather than a federal record.
- The by-group figures carry every caveat above, unequally, so a published 19.9-point spread between two groups in 2024 is wrong by an unknown amount.

³ Robert M. Groves, "Nonresponse Rates and Nonresponse Bias in Household Surveys," *Public Opinion Quarterly* 70, no. 5 (2006): 646-675, doi:10.1093/poq/nfl033.

⁴ Aram Hur and Christopher H. Achen, "Coding Voter Turnout Responses in the Current Population Survey," *Public Opinion Quarterly* 77, no. 4 (2013): 985-993, doi:10.1093/poq/nft042; summarised at <https://electionupdates.caltech.edu/2014/01/21/more-new-research-on-voter-turnout-hur-and-achen-in-poq-on-coding-voter-turnout-responses-in-the-cps/>.

⁵ Matthew K. Berent, Jon A. Krosnick and Arthur Lupia, "Measuring Voter Registration and Turnout in Surveys: Do Official Government Records Yield More Accurate Assessments?" *Public Opinion Quarterly* 80, no. 3 (2016): 597-621, doi:10.1093/poq/nfw021.

06 Extensions

None of these is answered above, and every one can be answered from the table the Sources section hands over.

- `cps_turnout.csv` carries `cps_rate` and `actual_rate` side by side with `over_report_pp` between them. Sort by `over_report_pp` and read the top and bottom five years. Do the worst years share anything?
- The same file has `vap_thousands` and `citizen_vap`. Compute `actual_votes` divided by each of them for every year since 1980, and see how far apart the two turnout rates run.
- Compare `pct_white_nh`, `pct_black`, `pct_asian` and `pct_hispanic` across the years present. Which group's reported turnout moved most between 2016 and 2020, and which moved least?
- Take `reported_voters` minus `actual_votes` for one year and divide it by the state populations you can find. How many states' worth of people are in that difference?
- **Stretch.** The Bureau publishes the same supplement for midterm years. Download two of them, set the reported voters against the ballots counted, and see whether the gap is larger in a midterm than in a presidential year.
- **Stretch.** Several states publish a voter file recording whether each registrant voted. Take one state's file and compare its turnout to the survey's figure for that state, which is a join on a much worse key than a year.

07 Sources

Data

- U.S. Census Bureau, Current Population Survey, *Voting and Registration Supplement*, Table A-1, November 1964–2024. <https://www.census.gov/data/tables/time-series/demo/voting-and-registration/voting-historical-time-series.html>
- Counted ballots: national presidential vote totals, 1964–2024, from Jason Timm, *Pre-electionResults* (<https://github.com/jaytimm/PresElectionResults>), which compiles the per-election encyclopedia articles that rest on Dave Leip's *Atlas of U.S. Presidential Elections* and, beneath that, the official state canvasses.

The data itself

Every figure above is drawn from this table. It is plain CSV, it opens in a spreadsheet, and it is yours to take further.

`cps_turnout.csv`

Academic literature

- Ansolabehere, S. and Hersh, E. (2012). "Validation: What Big Data Reveal About Survey Misreporting and the Real Electorate." *Political Analysis* 20(4): 437–459. <https://doi.org/10.1093/pan/mps023>
- Berent, M. K., Krosnick, J. A. and Lupia, A. (2016). "Measuring Voter Registration and Turnout in Surveys: Do Official Government Records Yield More Accurate Assessments?" *Public Opinion Quarterly* 80(3): 597–621. doi:10.1093/poq/nfw021
- Groves, R. M. (2006). "Nonresponse Rates and Nonresponse Bias in Household Surveys." *Public Opinion Quarterly* 70(5): 646–675. doi:10.1093/poq/nfl033

- Hur, A. and Achen, C. H. (2013). "Coding Voter Turnout Responses in the Current Population Survey." *Public Opinion Quarterly* 77(4): 985-993. doi:10.1093/poq/nft042; summarised at <https://electionupdates.caltech.edu/2014/01/21/more-new-research-on-voter-turnout-hur-and-achen-in-poq-on-coding-voter-turnout-responses-in-the-cps/>
- McDonald, M. P. and Popkin, S. L. (2001). "The Myth of the Vanishing Voter." *American Political Science Review* 95(4): 963-974. <https://doi.org/10.1017/S0003055400400134>
- U.S. Census Bureau, *Current Population Survey: Methodology* – the sample of about 60,000 households and how it is drawn. <https://www.census.gov/programs-surveys/cps/technical-documentation/methodology.html>

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original source, without any starting files. Please do the following and show your work.

THE SOURCE. U.S. Census Bureau, Current Population Survey, Voting and Registration Supplement, historical Table A-1. One Excel workbook, about 44 KB, published at

https://www2.census.gov/programs-surveys/cps/tables/time-series/voting-historical-time-series/hst_vote01.xlsx

No account or key is needed. This is a survey: tens of thousands of households are asked, after each November election, whether they voted, and nobody checks their answers. The sheet stacks several blocks; the one you want sits under the label "Total, Percent Voted", one row per election year, giving the voting-age population in thousands and the percent who said they voted on two bases – the total population, and citizens only.

WHAT TO BUILD.

1. For each presidential election year, the number of people who told the survey they voted. The table never prints that number; derive it as the voting-age population times the percent voted on the total basis. Derive the citizen voting-age population the same way, from the ratio of the two percentage columns.
2. Set those reported voters beside the ballots actually counted for president each year, from any official tally of national election returns, and compute the gap. Expect the survey to run high: people say they voted when they did not.

CHECK YOUR WORK. The copy this book used was downloaded in August 2026.

If your rebuild matches it, you should find:

- 16 presidential elections, 1964 through 2024
- in 2024, a voting-age population of 260,363 thousand and 154,395,259 reported voters, against 152,320,193 votes counted for president – a gap under one percentage point
- in 1992 the gap was near six points; it shrinks decade by decade and is nearly gone by 2020

The Bureau extends this table after every federal election and can revise old rows; if a number moved, the table was revised, not misread.